BoilerVec Eisenhower Matrix

Vector database built from scratch in Python, powering a live Purdue course search. MVP due November 24, 2026.

	Urgent	Not Urgent
Important	**Do (Your MVP)** • Brute-force index (ground truth) • HNSW insert and search • Neighbor-selection heuristic • Cosine and L2 distance • Recall@k benchmark harness • Benchmarks vs. hnswlib • Binary snapshot format • Write-ahead log with crash recovery • Fault-injection crash tests • HNSW vs. brute-force property tests • Upsert and query API endpoints • Purdue course data ingestion • React course search page • Live deployment with HTTPS • README with benchmark charts	**Schedule (For V2.0+)** • Metadata filters • Deletes (tombstones) • Vector compression (quantization) • pip-installable package • Compiled hot path (Numba or C) • 1M-vector benchmarks • Write-ahead log compaction • Concurrent reads during writes • Automatic catalog refresh each semester
Not Important	**Delegate (Use a library)** • Text embeddings (hosted API) • Vector math (NumPy) • Web framework (FastAPI) • Frontend build tooling (Vite) • HTTPS certificates (Caddy) • Containers (Docker Compose) • Benchmark datasets (ann-benchmarks) • Course catalog data (Purdue.io) • Rate limiting (slowapi) • Property-based testing (Hypothesis) • Test runner (pytest) • CI (GitHub Actions) • Charts (matplotlib) • Checksums (zlib CRC32)	**Delete (Cut it)** • RAG chat assistant • Multi-server sharding • Beating hnswlib on raw speed • User accounts and login • Saved searches and favorites • Multi-page frontend • GPU acceleration • Custom query language • Kubernetes deployment • Mobile app